Credit Card Default Prediction Using Machine Learning

YU LING SUN

October 01, 2025

Abstract

This study aims to predict the probability of default for credit card clients using machine learning techniques. Utilizing the "Default of Credit Card Clients Dataset" (30,000 samples), we performed data preprocessing, exploratory data analysis (EDA), baseline modeling, tree-based modeling, and cost-sensitive XGBoost optimization. Results show that Random Forest provides the strongest F1-score among baseline models, while the optimized XGBoost model significantly improves Recall, making it more suitable for financial risk management applications. By addressing class imbalance and enhancing financial interpretability through feature engineering, the study demonstrates how cost-sensitive learning and engineered features substantially improve default detection.

1 Introduction

Credit risk prediction allows financial institutions to adjust credit exposure, reduce losses, and improve portfolio stability. The dataset contains 30,000 credit card clients in Taiwan, with a default rate of approximately 22%, creating a class imbalance challenge that significantly impacts model sensitivity. Accurately identifying defaulting clients is critical, as missing high-risk cases produces severe financial consequences. This motivates the development of models that focus not only on accuracy but also on maximizing Recall (sensitivity to defaulters).

2 Methodology

2.1 Data Preprocessing

ID columns were removed and missing values checked. No missing entries were found. Because the target distribution is imbalanced, a stratified train-test split (70%–30%) was chosen to preserve proportionality. Initial exploratory analysis confirmed the imbalance (77.8% non-default vs. 22.2% default). This imbalance further supports the need for cost-sensitive learning in later modeling stages.

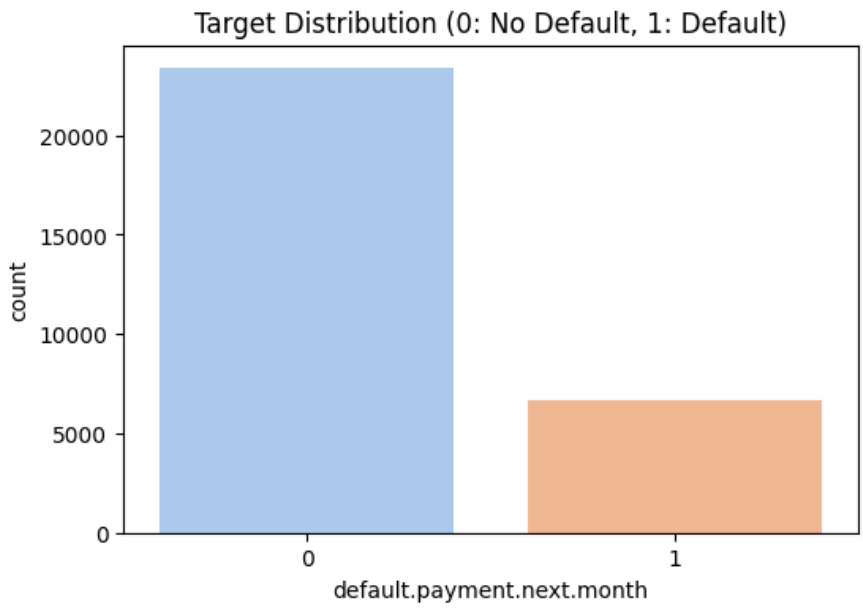

Figure 1: Target Variable Distribution (0 = No Default, 1 = Default)

2.2 Feature Engineering

To enhance interpretability and capture financial stress more accurately, three additional financial features were created:

- **Credit Utilization:** $BILL_AMT1/LIMIT_BAL$, capturing whether clients are approaching their credit limits.
- **Payment Ratio:** $PAY_AMT1/BILL_AMT2$, indicating repayment sufficiency.
- **Bill Growth:** $(BILL_AMT1 - BILL_AMT2)/BILL_AMT2$, detecting sudden increases in spending.

Exploratory visualizations revealed key financial behaviors. Clients with high repayment delays ($PAY_0 \geq 2$) exhibited much higher default rates, while lower credit limits tended to cluster with defaulters. These findings supported the inclusion of engineered features tailored to financial stress.

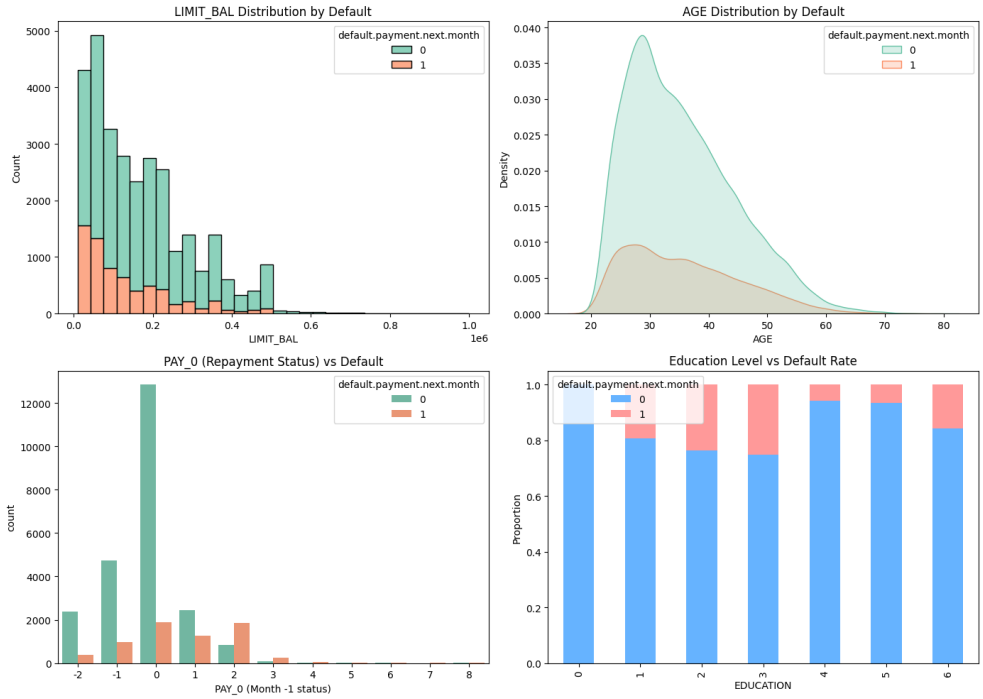

Figure 2: Exploratory Analysis of LIMIT_BAL, AGE, PAY_0, and EDUCATION

2.3 Modeling Strategy

We evaluated four models:

1. Logistic Regression (baseline linear model)
2. Decision Tree (baseline nonlinear model)
3. Random Forest optimized with Grid Search to maximize F1-score
4. XGBoost with feature engineering and cost-sensitive learning

Random Forest served as the primary ensemble model in the baseline phase. However, ensemble methods without cost adjustments still

tend to underperform when positive examples (defaulters) are scarce. Therefore, the study incorporated a cost-sensitive XGBoost model in later stages to prioritize Recall.

3 Experimental Results

3.1 Performance Metrics

The following table summarizes model performance on the test set:

Table 1: Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	0.8022	0.74	0.59	0.61
Decision Tree	0.7232	0.60	0.61	0.61
Random Forest	0.8200	0.75	0.65	**0.68**
XGBoost (Cost-Sensitive)	0.7600	0.67	**0.71**	0.68

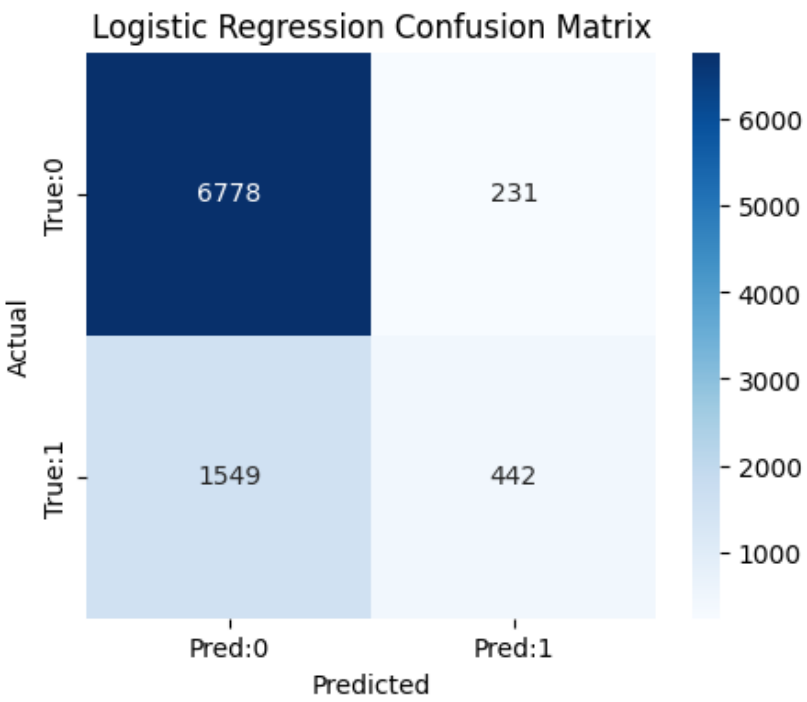

Figure 3: Confusion Matrix — Logistic Regression

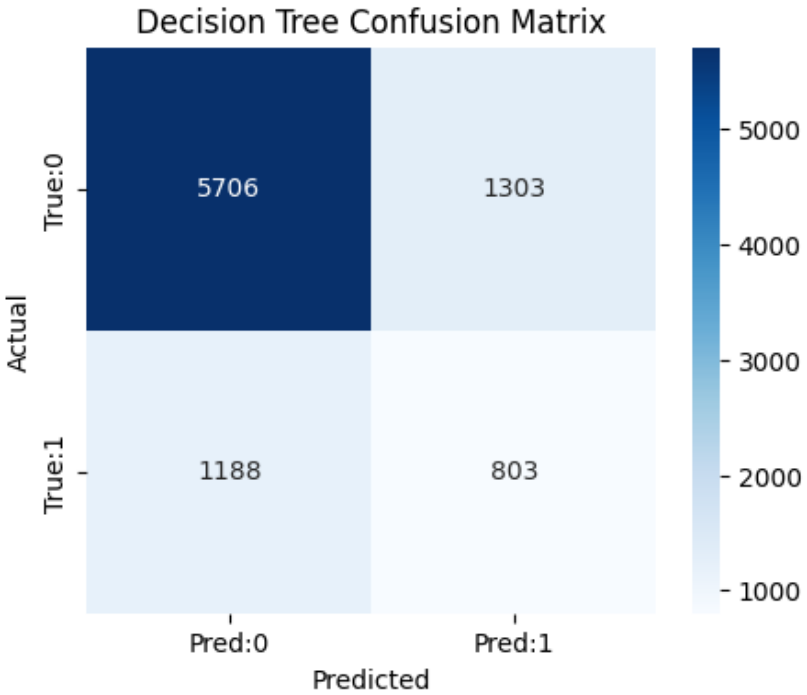

Figure 4: Confusion Matrix — Decision Tree

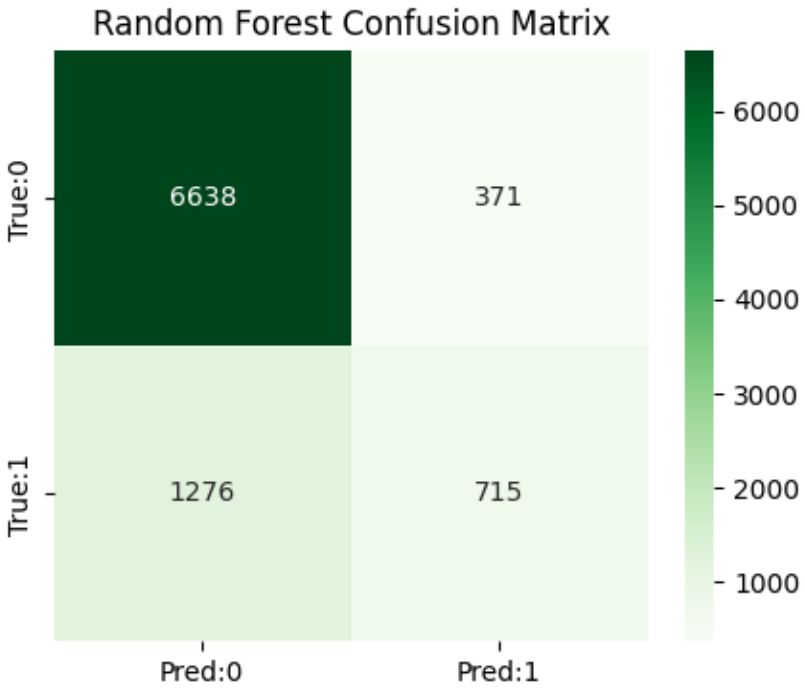

Figure 5: Confusion Matrix — Random Forest

The Random Forest model demonstrated strong overall performance and notably higher F1-score. However, its Recall for default cases remained lower than desired for real-world risk management contexts. This limitation motivated the next phase of optimization.

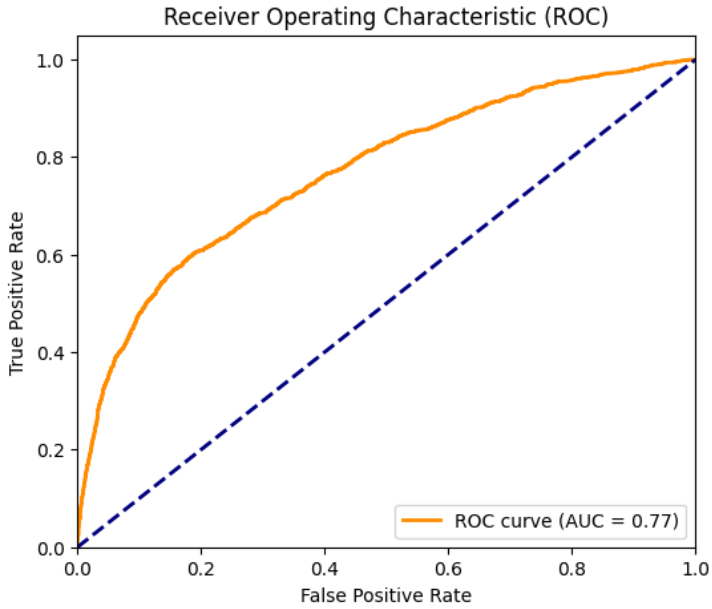

Figure 6: ROC Curve for Random Forest (AUC = 0.77)

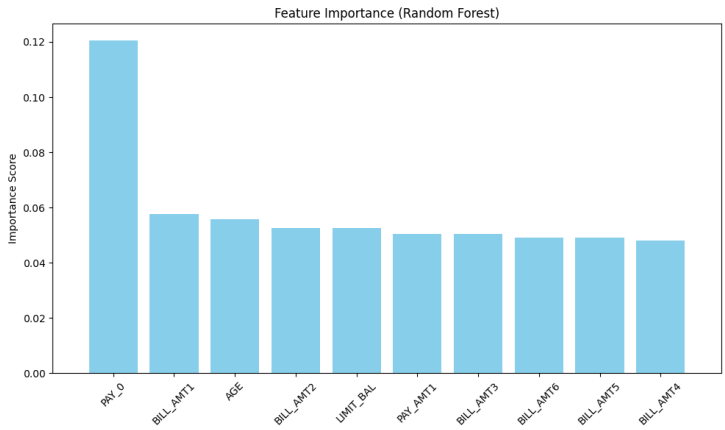

Figure 7: Random Forest Feature Importance Ranking

3.2 Advanced Model: Cost-Sensitive XGBoost

As an advanced extension of this study, we addressed the inherent class imbalance problem using Cost-Sensitive Learning. Standard classification models often prioritize overall accuracy, which can lead to poor sensitivity towards the minority class (defaulters). To overcome this limitation, we implemented an XGBoost model with

the scale_pos_weight hyperparameter set to approximately 3.5 (the ratio of negative to positive samples).

This configuration forces the algorithm to penalize "missed defaulters" (False Negatives) significantly more heavily than standard errors. By assigning a higher weight to the minority class during the gradient boosting process, the model shifts its decision boundary to become more risk-averse. As a result, Recall increased substantially, demonstrating that the model effectively prioritizes risk detection over simple accuracy|a crucial advantage in financial risk management where failing to identify a default is far more costly than a false alarm.

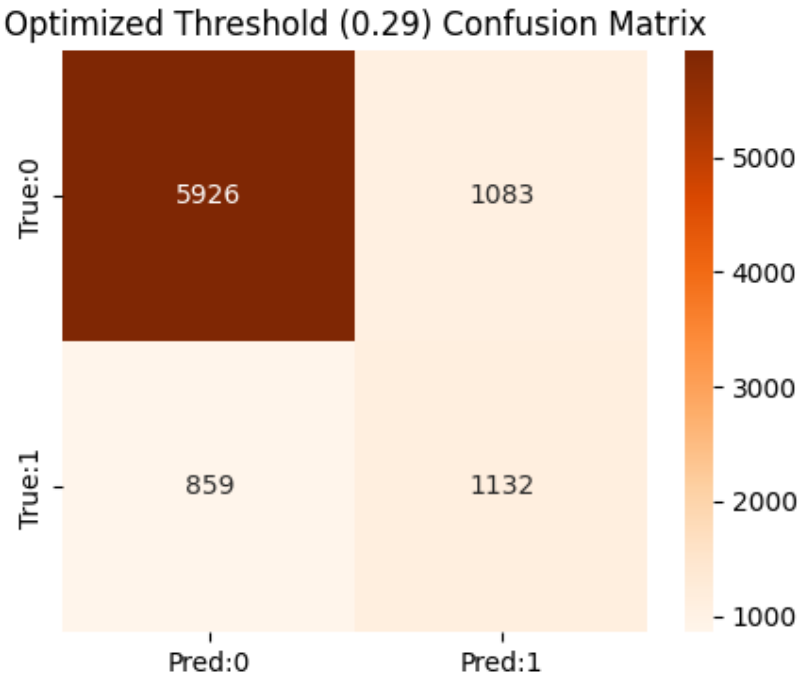

Figure 8: Confusion Matrix After Threshold Optimization

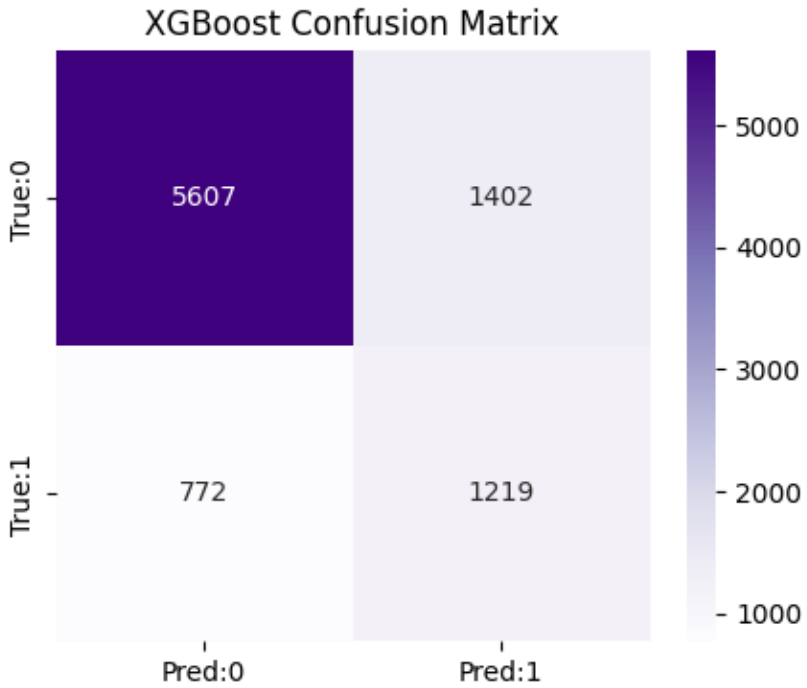

Figure 9: Confusion Matrix — XGBoost with Cost-Sensitive Learning

The XGBoost model produced a much larger number of true positives, confirming that cost-sensitive learning effectively mitigates the model's tendency to ignore minority-class samples.

4 Conclusion

This study compared baseline models, Random Forest, and a cost-sensitive XGBoost model for credit default prediction. Random Forest provided strong general performance, but XGBoost offered a substantial improvement in Recall, which is the most crucial metric in financial risk control. Feature engineering also increased model interpretability, with engineered variables such as utilization and payment ratio appearing among the top predictors. In real financial settings, better Recall translates into earlier detection of high-risk clients, reducing potential losses and improving credit portfolio stability.